

Course Title: Computer Networks

Course Code : CSE 204

Credits: 4(3|1)

Prerequisites : CSE 100

Co-Requisites:

School/Department: School of Computing and Information Technology

Course Description

The course provides general knowledge about computer networks such as the OSI model, peer to peer network, Client-Server network, LAN, MAN, WAN, protocols network, IP Address, network devices(Router, Switch, Hub, Wireless,...). In addition, the course also presents solutions for connecting network devices to deploy small computer network systems.

Course Objectives

Computer Network course aims to train students to obtain the necessary knowledge to:

- Understand the concepts, technologies, architecture of Computer Networks, and transmission of signal computers.
- Apply the architecture of OSI model, TCP/IP, the protocols and architecture Internet.
- Analyze and implement small computer network systems.

Course Intended Learning Outcomes

At the end of this course, students will be able to:

- CLO1. Explain the concepts related to Computer Networks (architecture of OSI, TCP/IP, the protocols and architecture Internet)
- CLO2. Apply the knowledge about LAN, function of each layer in the OSI, the protocols, network devices in designing and implementing LAN system.
- CLO3. Analyze and solve the problems related to computer networks.
- CLO4. Use some softwares to config LAN system.
- CLO5. Combine groups to work effectively.
- CLO6. Have a serious manner, professional working attitude. Attend full course and be on time.

Course Content

[Type here]

Chapter 1. Introduction to computer network

- 1.1. Exploring the functions of networking
- 1.2. Attack and Security network
- 1.3. OSI model

Chapter 2. Introduction to the TCP/IP

- 2.1. Introduction to the TCP/IP
- 2.2. Introduction to the IP Address(Ipv4, Ipv6)
- 2.3. TCP Three Way Handshake
- 2.4. Mapping L2 Address to L3 Address

Chapter 3. Subnetting

- 3.1. Introduction Subnetting
- 3.2. VLSM

Chapter 4. Introduction Ethernet LAN & Cable

- 4.1. LAN def – Ethernet cable - SW
- 4.2. PC – SW - PC
- 4.3. Routing & router (LANs connection)

Chapter 5. Routing

- 4.1. Static Route
- 4.2. Dynamic Route: RIP, IGRP
- 4.3. Classes of Routing Protocols
- Review for Midterm

Chapter 6. DHCP, NAT,.

- 6.1. How DHCP work, and DHCP configuration
- 6.2. What is NAT
- 6.3. Static NAT
- 6.4. Dynamic NAT

Chapter 7. Network Design

- 7.1. Analyzing
- 7.2. Designing
- 7.3. Implementing
- 7.4. Operating

Chapter 8. Virus - FTP

- 8.1. Virus, Worm, Trojan Horse & Backdoor
- 8.2. How FTP works

Review for Final

Course Materials¹ & Other Requirements

Textbook:

[1]. Tanenbaum, A. S. (2021). *Computer networks*. Upper Saddle River (NJ): Prentice Hall PTR.

Required Readings:

[2] Forouzan, B. A. (2010). *TCP/IP protocol suite*. Boston: McGraw-Hill.

[3] Kurose, J. F., & Ross, K. W. (2017). *Computer networking: A top-down approach*. New York: Addison-Wesley.

[4] Cisco Networking Academy. (2020). *Introduction to Networks 7.0*. Cisco Press.

Additional Readings: None

Other requirements (software, equipment, online resources, etc.): VMware software, Cisco Packet Tracer, Switch, Router.

Course Assessment ²

Assessment Type	Assessment Method	Description	Weight (%)		Aligned CLO
Formative	[1] Attendance	Attendance, Regularity, punctuality	35	10	CLO6
	[2] Practical Assignment on Computer	Lab 1: Configure MS LAN network, Using wireshark to catch packet. Lab 2: Network subnetting assignments Lab 3: Getting familiar with Packet Tracer Lab 4: configure static route, configure dynamic route Lab 5: Assignments about: VLAN, DHCP. Lab 6: Configure Static NAT, Dynamic NAT Lab 7: Sharing folder on windows		25	CLO2-5
Midterm	[3] Exam	Multiple choices questions.	20	20	CLO1-2
Final	[4] Exam	Multiple choices questions and written test.	45	45	CLO1-3

¹ Available at the EIU Library

² The sum of x, y, and z must be 100%;
 $\sum x_i = \sum y_i = \sum z_i = 100\%$

Course Schedule ³

Week 1, Lesson 1 4 hrs theory 4 hrs practice Aligned CLO:1,2	Chapter 1. Introduction to computer network Reading - Chapter 1 – Learning Materials [1] - Chapter 2 – Learning Materials [2] Discussion - Lecture slides Chapter 1 - Features of OSI model Practice: None
Week 2, Lesson 2 4 hrs theory 4 hrs practice Aligned CLO:1-3; 5-6	Chapter 2. Introduction to the TCP/IP Reading - Chapter 1,6 – Learning Materials [1] - Chapter 2 – Learning Materials [2] - Lecture slides Chapter 2 Discussion - Lecture slides Chapter 2 - Subnetting Practice - Configure MS LAN network, Using wireshark to catch packet. Assessment: [Practical Assignment on Computer]
Week 3, Lesson 3 4 hrs theory 4 hrs practice Aligned CLO:1-3; 5-6	Chapter 3. Subnetting Reading - Chapter 5 – Learning Materials [1] - Lecture slides Chapter 3 Discussion - Lecture slides Chapter 3 - TCP protocol Practice Network subnetting, Config IP for computer Assessment: [Practical Assignment on Computer]
Week 4, Lesson 4 4 hrs theory	Chapter 4. Introduction Ethernet LAN & Cable Reading Chapter 2,3,4 – Learning Materials [1]

³ Students need two hours of self-study for each hour of classroom instruction.

4 hrs practice Aligned CLO:1-3; 5-6	- Lecture slides Chapter 4 Discussion - Lecture slides Chapter 4 - LAN devices, Switch, VLAN, routing protocols Practice - Packet Tracer. Assessment: [Practical Assignment on Computer]
Week 5, Lesson 5 4 hrs theory 4 hrs practice Aligned CLO:1-3; 5-6	Chapter 5. Routing Reading - Chapter 5 – Learning Materials [1] - Lecture slides Chapter 5 Discussion - Lecture slides Chapter 5 - Compare routing protocols Practice - Configure static route, configure dynamic route Assessment: [Practical Assignment on Computer]
Week 6, Lesson 6 4 hrs theory 4 hrs practice Aligned CLO:1-3; 5-6	Chapter 6. DHCP& NAT Reading - Chapter 5 – Learning Materials [1] - Chapter 2 – Learning Materials [2] - Lecture slides Chapter 6 Discussion - Lecture slides Chapter 6 - Explain NAT, DHCP service Practice - Config VLAN, DHCP, Static Routing, Dynamic Routing (RIP) - Sharing folder on windows - Config remote desktop Assessment: [Practical Assignment on Computer]
Week , Lesson 7	Chapter 7. Network Design Reading

4 hrs theory 4 hrs practice Aligned CLO:1-3; 5-6	- Chapter 4,5,8 – Learning Materials [1] - Lecture slides Chapter 7 Discussion - Lecture slides Chapter 7 - Design computer network Practice - Config static NAT, dynamic NAT Assessment: [Practical Assignment on Computer]
Week 8, Lesson 8 4 hrs theory 4 hrs practice Aligned CLO:1-3; 5-6	Chapter 8. Virus - FTP Reading - Chapter 7, 8 – Learning Materials [1] - Lecture slides Chapter 8 Discussion - Lecture slides Chapter 8 - Virus - FTP Practice - Sharing folder on windows with ftp server application (Filezilla Server and Filezilla Client). Assessment: [Practical Assignment on Computer]
Week 9, Lesson 9 2 hrs practice Aligned CLO:5-6	Practice Question & Answer Session and Discussion Packet Tracer, subnetting Assessment: [None]

Course Policies

- If a student is absent $\geq 20\%$, then the student will not be allowed to take the final examination. (Absent $\geq 4 \Rightarrow$ prohibited; Absent=3(-75%); Absent=2(-50%); Absent=1(-25%)).
- The practical exercises/assignments score is the average of all practical exercises/assignments.
- Copying from others will carry minus 100% of the points.

Ho Chi Minh, / / 2025

Dean**Head of department****Lecturer****Narayan C. Debnath****Huynh Tan Phuoc****Huynh Tan Phuoc**



Appendix 1. Rubrics ⁴

Rubric [1]

Criteria	CLOs	Weight (%)	Failure	Satisfactory	Good	Excellent	Point
			25% of the maximum point	Medium 50% of the maximum points	Good 75% of the maximum points	Exceclent 100% of the maximum points	
Attendance, Punctuality in theory & laboratory classes	CLO6	100	Absent=3 (25%)	Absent=2(50%)	Absent=1(75%)	Absent=0(100%)	100
Total:							100

Rubric [2]

Criteria	CLOs	Weight (%)	Failure	Satisfactory	Good	Excellent	Point
			0-49.9	50-64.9	65-79.9	80-100	
Apply the knowledge about about LAN, function of each layer in the OSI, the protocols	CLO 2	40	Have thinking but not completed any problems.	Finish at least 50% of assignments. (50%-64.9%)	Finish all the assignments with exactly method but have some bugs that lead to incorrect result	Complete all the assignment with right results.	40
Analyze the solutions for the LAB assignments	CLO 3	20	Have thinking but not completed any problems.	Finish at least 50% of assignments. (50%-64.9%)	Finish all the assignments with exactly method but have some bugs that lead to incorrect result	Complete all the assignment with right results.	20
Use tools and configurations for the LAB assignments	CLO 4	20	Have using the tools but not well	Use the tools finish at least 50% of assignments. (50%-64.9%)	Use the tools finish all the assignments but have some	Use the tools complete all the assignment with right results.	20

⁴ The rubrics depend on each school's policies, concentrations and courses
[Type here]

					bugs that lead to incorrect result		
Combine groups to solve for the LAB assignments.	CLO 5	20	Have discussion but not completed any problems.	Every member participates in discussion and work.	All members participated in the discussion and resolved well.	All members actively participated in discussions and resolved very well.	20
Total:							100

Rubric [3]

Criteria	CLOs	Weight (%)	Failure	Satisfactory	Good	Excellent	Point
			0-49.9	50-64.9	65-79.9	80-100	
Accuracy	Midterm exam: CLO 1,2	100	Answers less than 13 multiple choice questions correctly	Answers 13-16 multiple choice questions correctly	Answers 17-19 multiple choice questions correctly	Answers >=20 multiple choice questions correctly	100
Total:							

Rubric [4]

Criteria	CLO	Weight (%)	Failure	Satisfactory	Good	Excellent	Point
			0-49.9	50-64.9	65-79.9	80-100	
Accuracy	Final exam: CLO 1,2	40	Answers less than 50% multiple choice questions correctly	Answers multiple choice questions correctly (50%-64.9%)	Answers multiple choice questions correctly (65%-79.9%)	Answers multiple choice questions correctly (80%-100%)	40
Apply	CLO 2	50	Have thinking but not completed any problems.	Finish at least 50% of questions. (50%-64.9%)	Finish all the questions with exactly method but have some bugs that lead to incorrect result	Complete all the questions with right results.	50

Analysis	CLO 3	10	Lacks analysis or evaluation of the issues and events beyond stating vague, irrelevant, and/or inaccurate facts	Lacks analysis or evaluation of the issues and events beyond stating accurate, relevant facts	Analyzes or evaluates issues and events, but not in any depth	Analyzes, evaluates, compares, and/or contrasts issues and events with depth.	10
						Total:	100

Appendix 2. Mapping Between CLO And PLO

CLO	PLO
	Computer Networks and Data Communications
1. Explain the concepts related to Computer Networks (architecture of OSI, TCP/IP, the protocols and architecture Internet	PLO2.2
2. Apply the knowledge about LAN, function of each layer in the OSI, the protocols, network devices in designing and implementing LAN system.	PLO2.2
3. Analyze and solve the problems related to computer networks.	PLO2.3
4. Use some softwares to config LAN system.	PLO6.1, PLO 6.2
5. Combine groups to work effectively.	PLO5.1
6. Have a serious maner, professional working attitude. Attend full course and be on time.	PLO8.1

///TEST BLUEPRINT is uploaded on Moodle///

Midterm Examination Blueprint

Final Examination Blueprint

Assessment Content	CLO	Remember	Understand	Apply	Analyze	Evaluate	Create	Total ques.	Points
I.Multiple Choice Question									
Chapter 4: LANs connection, divide subnet	2			5				5	10
Chapter 5: Routing	2			5				5	10
Chapter 6: DHCP& NAT	2			2				2	4
Chapter 7: Network Design				6				6	12
Chapter 8: Virus - FTP	2			2				2	4
II. Written questions (exercise)									
Chapter 4-8: Apply IP to divide subnet VLSM, VLAN, Routing, DHCP,NAT service	2			3					50
Analyze design computer network	3				1				10
Total:								25	100

